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FACULTY OF ENGINEERING & COMPUTER SCIENCE
DEPARTMENT OF COMPUTER SCIENCE &
INFORMATION SYSTEMS
CS323 DATABASE SYSTEMS

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Database Project Presentation and Submission Guidelines

The purpose of the presentation is for each group to demonstrate that they can move from **database analysis and design to a working application**. The assessment, therefore, focuses not only on the application's appearance but also on the quality of the underlying database and the students' understanding of their design decisions.

1. Presentation Requirements

Each group will have a **maximum presentation time of 15 minutes**, structured as follows:

Component	Time	What the Group Should Do
1. Introduction	1 minute	Introduce the project, problem being solved, target users, and group members.
2. 3-Minute Project Video	3 minutes	Play a professionally prepared video showing the problem, database design, application features, and key functionality.
3. Database Design Explanation	3 minutes	Explain the ER diagram, entities, relationships, cardinalities, primary/foreign keys, and normalization.
4. Live Application Demo	6 minutes	Demonstrate the working application and show how it interacts with the database.
5. Questions & Answers	2 minutes	Respond to questions from the lecturer/assessors.
Total	15 minutes	

Important Rule

The 3-minute video is not a substitute for the live demonstration.

The video should provide an overview of the project, while the live demonstration should prove that the application actually works.

2. Guidelines for the 3-Minute Video

Each group must prepare a **maximum 3-minute video**.

The video should be structured approximately as follows:

0:00–0:30 — Problem Introduction

Briefly explain:

- What real-life problem are you solving?
- Who experiences the problem?
- Why is a database solution appropriate?
- Who are the intended users?

Example:

"Our project is a University Hostel Management System designed to address challenges associated with room allocation, student payments, occupancy tracking, and maintenance requests."

0:30–1:00 — Database Design

Show:

- ER diagram
- Major entities
- Relationships
- Primary and foreign keys
- Brief indication of normalization

The ER diagram should be clearly visible in the video.

1:00–2:30 — Application Demonstration

Show short clips of the major application features, such as:

- Login
- Dashboard
- Adding records
- Searching records
- Updating records
- Deleting records
- Generating reports
- Transaction processing
- Relevant business functionality

Do **not** spend the entire video showing login screens or attractive interfaces.

2:30–3:00 — Conclusion

Show:

- The completed application
- One or two important reports/results
- The problem solved
- Key benefits of the system

Video Quality Requirements

The video should:

- Be **no longer than 3 minutes**
 - Have clear narration
 - Have readable text
 - Show the actual system developed by the group
 - Avoid unnecessary animations
 - Clearly demonstrate database functionality
 - Include the project title and group number
 - Include names of group members at the beginning or end
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3. Live Application Demonstration

The live demonstration is one of the most important components of the presentation.

The group should demonstrate the **actual working application**, not screenshots or a PowerPoint simulation.

A. Login and User Management

Demonstrate, where applicable:

- User login
- Different user roles
- Authentication
- Role-based access
- Logout

For example:

Administrator → Full access

Staff → Limited access

Customer/Student → View and transaction access

B. CRUD Operations

Demonstrate at least one complete example of:

Create → Read → Update → Delete

For example, a hostel system could demonstrate:

1. Add a new student
2. View student information
3. Update student information
4. Delete/deactivate the student

The application should actually update the underlying database.

C. Search and Filtering

Demonstrate meaningful searches.

Examples:

- Find all students in a particular hostel
 - Find vehicles within a particular price range
 - Find patients treated by a particular doctor
 - Find products supplied by a particular supplier
 - Find unpaid hotel reservations
-

D. Database Transactions

Where appropriate, demonstrate a transaction involving multiple tables.

For example:

Car Dealership

Customer

↓

Vehicle Purchase

↓

Payment

↓

Vehicle Inventory Updated

Students should explain how the database maintains consistency during the transaction.

E. Reports

Demonstrate at least **two meaningful reports** generated from the database.

Examples:

- Monthly sales report
- Outstanding payments
- Hostel occupancy report
- Hospital appointments
- Low-stock medicines
- Vehicle sales by month
- Waste collection statistics

Reports should be generated from actual database data.

4. Database Demonstration

During the presentation, groups should briefly demonstrate that the application is backed by a properly designed relational database.

They should be prepared to show:

Database Structure

- Tables
- Primary keys
- Foreign keys
- Relationships
- Constraints
- Appropriate data types

SQL

The group should be prepared to show examples of:

- CREATE TABLE
- INSERT
- UPDATE

- DELETE
- SELECT
- JOIN
- Aggregate functions
- Subqueries
- Views
- Stored procedures/functions
- Triggers

The lecturer may ask students to execute an SQL query **during the presentation**.

This helps establish that students understand the database rather than simply using an application developed by someone else.

5. Questions the Group Should Be Prepared to Answer

Every member should understand the entire project.

Possible questions include:

ER Design

1. Why did you identify this as an entity?
2. What is the primary key of this table?
3. Why is this relationship one-to-many?
4. Why did you create an associative entity?
5. What are the business rules governing this relationship?

Normalization

6. Why did you normalize the database?
7. What was the original unnormalized structure?
8. How did you achieve 1NF, 2NF and 3NF?
9. What problems would occur if the database were not normalized?

SQL

10. Why did you choose this data type?
11. Why is this attribute a foreign key?
12. What does this query return?

13. Why did you use a JOIN?
14. Why did you create this view?
15. What business rule is implemented by the trigger?

Application

16. How does the application connect to the database?
17. What happens when invalid data is entered?
18. How is authentication implemented?
19. How do you prevent duplicate records?
20. What happens if a database transaction fails?

6. Group Project Document

Each group must submit **one comprehensive project report**.

The document should demonstrate the complete development process from **problem identification** → **database design** → **implementation** → **application development** → **testing**.

Recommended Report Structure

Chapter/Section	Content
Cover Page	Project title, group number, student names/IDs, course, lecturer, date
Table of Contents	Automatically generated contents
List of Figures	ER diagrams, screenshots, architecture diagrams, etc.
List of Tables	Database tables and other report tables
1. Introduction	Background, problem statement, motivation, objectives, scope
2. Requirements Analysis	Stakeholders, users, functional requirements, non-functional requirements, assumptions
3. System Analysis	Existing system, problems with existing system, proposed solution, business rules
4. Database Design	ER diagram, entities, attributes, relationships, cardinalities
5. Database Implementation	Relational schema, normalization, tables, keys, constraints, indexes
6. SQL Implementation	DDL, DML, queries, views, procedures, functions, triggers
7. Application Design	Architecture, user interface design, modules, navigation

8. Application Implementation	Description and screenshots of implemented functionality
9. Testing	Test strategy, test cases, expected results, actual results
10. Security and Data Integrity	Authentication, authorization, constraints, validation, backup considerations
11. Challenges and Solutions	Problems encountered and how they were resolved
12. Conclusion	Summary, achievements, limitations, future improvements
References	Books, articles, websites, documentation
Appendices	SQL scripts, additional screenshots, sample data, etc.

7. Screenshots Required in the Report

Screenshots should provide **evidence that the system was actually implemented**.

Do not simply include dozens of screenshots. Each screenshot should have a purpose and a caption.

Database Screenshots

Include screenshots showing:

- Database/schema
- Tables
- Table structures
- Primary keys
- Foreign keys
- Relationships
- Sample records
- Important SQL queries
- Query results
- Views
- Stored procedures/functions
- Triggers where applicable

Application Screenshots

Include screenshots of:

1. Login page
2. Main dashboard
3. Main data-entry form
4. Record listing
5. Search/filter functionality
6. Update functionality
7. Delete/deactivation functionality
8. Important transaction screen
9. Report generation
10. Error/validation handling
11. User/role management, if applicable
12. Other important project-specific features

Every screenshot should have a caption.

For example:

Figure 8.3: Student Registration Interface

Then explain what the screenshot demonstrates.

8. Evidence of Database-Application Integration

This is particularly important.

The report should demonstrate that the application is **actually connected to the database**.

Students should include:

Example 1 — Application Input

Screenshot showing:

Student Registration Form

Example 2 — Database Record

Screenshot showing the corresponding record stored in the database.

Example 3 — Application Output

Screenshot showing the record retrieved and displayed by the application.

This creates a clear chain:

User → Application → Database → Application → User

9. SQL Appendix

Each group should submit the complete SQL script used to create and populate the database.

The script should include:

1. CREATE DATABASE / schema
2. CREATE TABLE
3. PRIMARY KEY constraints
4. FOREIGN KEY constraints
5. UNIQUE constraints
6. CHECK constraints
7. INSERT statements
8. UPDATE statements
9. DELETE statements
10. SELECT queries
11. JOIN queries
12. Aggregate queries
13. Subqueries
14. Views
15. Stored procedures/functions
16. Triggers

The SQL should be organized and commented.

10. Testing Section

Students should not simply state:

"The system was tested and works."

They should provide **evidence of testing**.

A test table such as the following should be included:

Test ID	Feature	Test Input	Expected Result	Actual Result	Status
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T01	Login	Valid username/password	User successfully logs in	User logged in	Pass
T02	Login	Invalid password	Error message displayed	Error displayed	Pass
T03	Add Student	Valid student data	Student saved	Student saved	Pass
T04	Add Student	Duplicate student ID	Record rejected	Record rejected	Pass
T05	Search	Existing student ID	Student returned	Student returned	Pass
T06	Delete	Existing record	Record removed/deactivated	Record removed	Pass

Groups should include screenshots of selected test cases as evidence.

11. Recommended Submission Package

Each group should submit the following:

A. Project Report

PDF format

Contains:

- Complete documentation
- ER diagrams
- Database design
- SQL explanations
- Application architecture
- Screenshots
- Testing
- References

B. SQL Files

Include:

- Database creation script
- Table creation script

- Sample data
- Queries
- Views
- Procedures/functions
- Triggers

C. Application Source Code

Submit the complete source code.

The project should be organized so that the lecturer can identify:

Project/

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└─ Documentation/

└─ Database/

|

└─ create_database.sql

|

└─ create_tables.sql

|

└─ insert_data.sql

|

└─ queries.sql

|

└─ views.sql

|

└─ procedures.sql

|

└─ triggers.sql

|

└─ Application/

|

└─ source code

|

└─ configuration files

|

└─ Screenshots/

|

└─ Video/

|

└─ project_demo.mp4

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└─ README.md

D. 3-Minute Video

Submit the final video separately.

E. README File

The README should explain:

- Project name
 - Group members
 - Technologies used
 - Database management system used
 - Programming language/framework
 - How to install the application
 - How to create the database
 - How to populate the database
 - How to run the application
 - Test account credentials, if applicable
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12. Presentation Rules

To ensure fairness between groups:

1. **Every group member must participate in the presentation.**
2. Every member must understand the database and application.
3. The video must not exceed **3 minutes**.
4. The live demonstration must use the group's actual application.
5. Pre-recorded demonstrations cannot replace the live demo.
6. Students may use slides, but slides should not dominate the presentation.
7. The database must be accessible during the demonstration.
8. The lecturer may request an SQL query to be executed without prior preparation.
9. The lecturer may ask any group member to explain any part of the project.
10. Screenshots in the report must correspond to the actual system demonstrated.
11. Groups should prepare a backup copy of the application and database.
12. The presentation should focus on **technical implementation and problem solving**, rather than simply describing the project.

13.Assessment Structure

A **100-mark** assessment can be structured as follows:

Component	Marks
Problem Definition & Requirements	8
ER Diagram & Conceptual Design	12
Relational Design & Normalization	12
SQL Implementation	15
Database Integrity & Advanced Features	8
Application Functionality	15
User Interface & Usability	5
Testing & Validation	5
Project Documentation & Screenshots	8
3-Minute Video	4
Live Presentation & Demonstration	5
Questions & Technical Understanding	3
Total	100

Key principle

The project would be assessed as a **database application**, not merely as an application with a database attached.

Students should therefore be able to demonstrate the complete chain:

Real-life problem → Requirements → Business Rules → ER Diagram → Relational Schema → Normalization → SQL Database → Application → Testing → Working Solution.